

Luis Lara

Berkeley, CA | P: 1 (510)-253-3189 | luis_lara@berkeley.edu

EDUCATION

University of California, Berkeley

Berkeley, CA

B.S in Aerospace Engineering, Minor in EECS

Expected Grad: Dec 2028

Relevant Coursework: Solid Mechanics, Python, MATLAB, C++, Electricity and Magnetism, Circuits

Regents' and Chancellor's Scholar: UC Berkeley's merit-based award, granted to the top 1% of entering students

WORK EXPERIENCE

Lawrence Berkeley National Laboratory

Berkeley, CA

Biomechanics Research Intern

Jun 2025 – Aug 2025

- Conducted **biomechanics data analysis** using **Python** to study the **high-speed motion of sharpshooter** insects, generating **insights relevant to bio-inspired robotic mechanisms** for researchers at **Georgia Tech Robotics**.
- Developed and refined detailed **3D models of sharpshooter insects** using **MeshLab, Fusion 360, and Blender** to support scientific **analysis of complex motion systems**.
- **Led a team of 3** in developing a **VR visualization system** that allowed better **analysis of 3D sharpshooter models**.
- Conducted experiments at **ALS Beamline 8.3.2** to collect high-resolution scan data for **biomechanics analysis**.

Lawrence Berkeley National Laboratory

Berkeley, CA

BLDAP (Berkeley Lab Directors Apprenticeship Program)

Jun 2024 – May 2025

- Used **NumPy, Pandas, and Matplotlib** to analyze and visualize multiple **datasets across research areas including biofuel plant systems and astronomy**.
- Programmed **Arduino sensors and actuators in C++** to support a **solar-powered, energy-efficient house model**.
- Designed CAD models in **Fusion 360** and applied **3D printing** skills to develop components for an eco-efficient house
- **Taught** a class of **30+** elementary school students introductory coding with **Scratch** and 3D modeling with **Tinkercad**.

Experts Lumber Structures Corporation

Hayward, CA

Carpenter

Jun 2019 – Present

- Contributed to large-scale construction and fabrication projects, including **houses, fences, barns, and concrete work**
- Supported project **planning and material management** for jobs with budgets up to **\$1,500**.
- Performed **hands-on** construction work, including **long hours in fast-paced job-site environments**.
- **Extensive experience** operating power tools, including **drills, saws, nail guns, and concrete mixers** across residential construction projects.

Santa Clara University

Santa Clara, CA

SEEDs (Spring Engineering Education Days)

Mar 2024 - 2025

- Selected as **1 of 50 students from 500 applicants** to participate in hands-on engineering projects.
- Applied fundamentals of aerodynamics and fluid mechanics to **design** and play with a **lightweight plastic frisbee**.
- Built an **Arduino-based LED circuit**, applying fundamentals of electrical circuits, wiring, and basic embedded programming.
- Collaborated with a group to **design and construct** a bridge prototype capable of supporting a **75-lb payload**

ACTIVITIES/PROJECTS

STAR (Space Technologies And Rocketry)

Sep 2025 – Dec 2025

Propulsion General Member

- Worked on the **combustion chamber and injector system** to support rocket propulsion performance, focusing on proper **geometry, fit, and manufacturability**.
- Considered key propulsion factors such as **fuel/oxidizer flow, injector spray behavior, pressure, heat, and chemical combustion** effects when supporting design decisions.

Skills

Software: SolidWorks, Fusion 360, Onshape, Ansys, Python, C++, MATLAB, SQL, GitHub

Engineering: CAD, GD&T, FEA, soldering, 3D printing